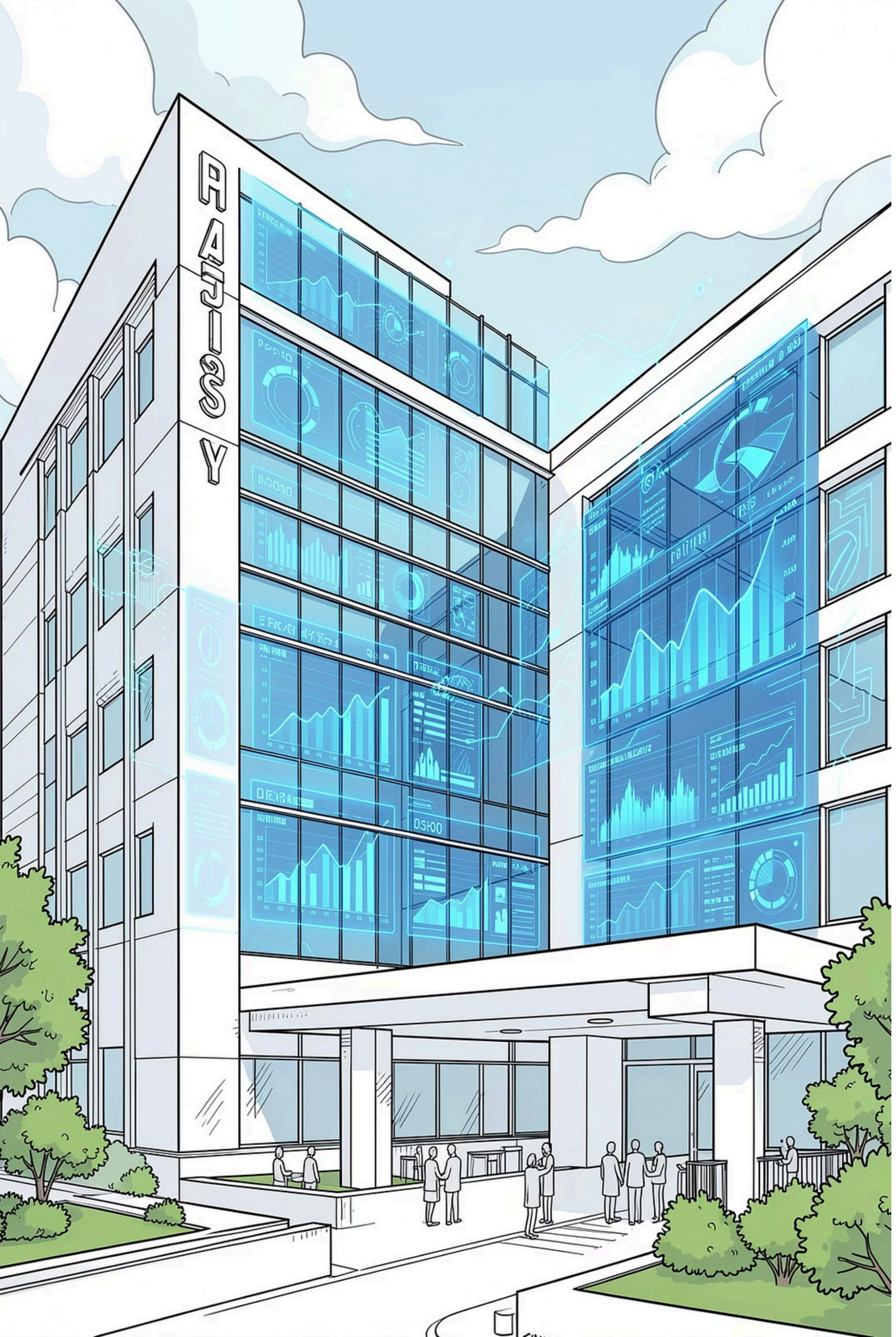

Hospital Performance & Patient Analytics

End-to-End Data Analytics Project — Transforming raw healthcare data into executive-level insights using SQL, Python, and Power BI.

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Tools Used: SQL · Python · Power BI · Excel · GitHub

Business Problem & Project Objectives

The Challenge

Hospital management lacks a centralized analytics solution to monitor patient admissions, billing performance, insurance providers, medical conditions, and operational efficiency. Without a unified data platform, leadership teams rely on fragmented reports and manual processes — slowing decision-making, obscuring revenue trends, and limiting the ability to identify operational bottlenecks. This project addresses that gap by building an end-to-end analytics pipeline that transforms raw healthcare data into structured, actionable intelligence.

> Healthcare organizations that leverage data analytics report up to 30% improvement in operational efficiency and measurable reductions in patient length of stay.

Project Objectives

Analyze Patient Demographics

Understand age distribution, gender ratios, and patient profiles across hospitals.

Monitor Admissions

Track admission types, volumes, and trends by hospital and time period.

Evaluate Billing Performance

Assess revenue generation, average billing, and cost variation by condition.

Analyze Insurance Providers

Identify top contributors to hospital revenue and coverage patterns.

Build an Executive Dashboard

Design an interactive Power BI dashboard for real-time monitoring.

Support Data-Driven Decisions

Deliver insights that enable hospital leadership to act with confidence.

Dataset Overview & Project Workflow

Dataset: Healthcare Dataset (Kaggle)

The project is built on a structured healthcare dataset comprising **15 columns** across four key information domains. The dataset covers patient demographics, hospital identifiers, billing records, insurance provider details, and diagnosed medical conditions — providing a comprehensive foundation for multi-dimensional analysis.

Patient Info

Age, Gender, Admission Type

Hospital Info

Hospital Name, Region

Billing Info

Cost, Insurance Claim

Medical Conditions

Diagnosis, Length of Stay

End-to-End Project Workflow

This project follows a structured six-phase analytics workflow — from initial business understanding through to the delivery of business insights. Each phase builds on the previous, ensuring data quality, analytical rigor, and business relevance at every step.

01

Business Understanding

Define KPIs, stakeholder requirements, and analytical scope.

02

Data Cleaning

Handle missing values, duplicates, and data type inconsistencies.

03

SQL Analysis

Query, aggregate, and transform data for business questions.

04

Python EDA

Explore distributions, correlations, and detect outliers.

05

Power BI Dashboard

Build interactive visualizations and executive reports.

06

Business Insights

Translate findings into actionable recommendations.

SQL & Python Analysis

Two complementary analytical engines were employed to extract, clean, explore, and interpret the healthcare dataset. SQL handled structured querying and aggregation, while Python enabled deeper statistical exploration and feature engineering.

SQL Analysis

SQL served as the primary tool for data extraction, transformation, and aggregation. Complex queries were written to answer specific business questions, compute KPIs, and prepare clean datasets for downstream visualization.

Data Cleaning

Removed duplicates, standardized categorical values, and handled NULL entries across 15 columns.

Aggregations & GROUP BY

Computed total revenue, patient counts, and averages segmented by hospital, condition, and insurance provider.

CASE Statements

Categorized admissions by urgency, age groups, and billing tiers for comparative analysis.

Window Functions

Ranked hospitals by revenue, calculated running totals, and computed percentiles for length-of-stay analysis.

Business Queries

Answered 10+ stakeholder questions including top-performing hospitals, highest-cost conditions, and insurance coverage gaps.

Python Analysis

Python (Pandas, NumPy, Matplotlib, Seaborn) was used for exploratory data analysis, statistical profiling, and visual discovery. Python enabled deeper investigation of patterns that SQL alone could not reveal.

Data Cleaning

Applied programmatic cleaning pipelines — imputation, type casting, and outlier capping — ensuring a reliable analytical base.

Feature Engineering

Created derived variables such as age brackets, billing categories, and admission urgency flags to enrich analysis.

Exploratory Data Analysis

Generated distribution plots, histograms, and bar charts to understand variable behavior across the dataset.

Correlation Analysis

Built correlation matrices to identify relationships between billing amounts, length of stay, and patient demographics.

Outlier Detection

Used IQR and Z-score methods to flag anomalous billing records and unusually long hospital stays for review.

Power BI Executive Dashboard

Dashboard Architecture

The Power BI dashboard was designed to serve hospital executives, operations managers, and financial analysts. It consolidates all key performance indicators and visualizations into a single interactive interface, enabling real-time monitoring of patient flow, revenue, and operational metrics. The dashboard uses Power Query for data transformation and DAX measures for dynamic KPI calculations.

Total Patients

Aggregate patient count across all hospitals and time periods.

Total Revenue

Sum of all billing amounts, segmented by hospital and provider.

Avg. Billing

Mean billing per admission, broken down by medical condition.

Avg. Age

Demographic profile of the patient population.

Avg. Length of Stay

Operational efficiency metric segmented by hospital.

Visual Components

Monthly Admissions Trend

Time-series line chart tracking patient volume month-over-month to identify seasonal patterns and peak periods.

Hospital Revenue Breakdown

Clustered bar chart comparing total revenue generated by each hospital, highlighting top and underperforming facilities.

Medical Conditions Distribution

Donut and bar charts showing the frequency of diagnoses, enabling resource planning for high-volume conditions.

Insurance Provider Analysis

Horizontal bar chart ranking insurance providers by patient coverage and total revenue contribution.

Age Group Segmentation

Pyramid or column chart segmenting patients by age bracket to support demographic and capacity planning.

Interactive Filters

Slicers for hospital, admission type, insurance provider, and date range enable drill-down exploration.

Key Business Insights

Six high-impact findings emerged from the analysis, each directly actionable for hospital leadership and operations teams.

1

Emergency Admissions Dominate

Emergency admissions represent the largest share of hospital visits, indicating a critical need for dedicated emergency capacity, rapid triage protocols, and surge staffing models to maintain quality of care during peak demand periods.

2

Revenue Concentrated in Few Hospitals

A small number of hospitals generate the majority of total revenue, suggesting potential network optimization opportunities and the need to investigate what differentiates high-performing facilities from lower-revenue counterparts.

3

Billing Varies by Medical Condition

Average billing amounts differ significantly across medical conditions, reflecting variation in treatment complexity, resource utilization, and length of stay — a key input for cost modeling and pricing strategy.

4

Insurance Providers Contribute Unequally

Revenue contribution varies widely across insurance providers. Strengthening partnerships with top-contributing insurers and expanding coverage agreements could directly improve hospital financial performance.

5

Middle-Aged Patients Drive Volume

Patients aged 35–54 account for the highest admission volumes, a demographic likely driven by chronic condition management and occupational health needs — a target segment for preventive care programs.

6

Length of Stay Differs Across Hospitals

Patient stay duration varies significantly between hospitals, even for similar conditions. This variation signals opportunities to standardize care pathways, reduce unnecessary stays, and improve bed turnover rates.

Strategic Recommendations

Based on the analytical findings, six targeted recommendations are proposed to improve operational efficiency, financial performance, and patient outcomes across the hospital network.

Optimize Staffing During Peak Admissions

Deploy predictive staffing models aligned with monthly admission trends, particularly for emergency departments. Use historical data to forecast surge periods and pre-schedule additional clinical and support staff to maintain care quality and reduce wait times.

Strengthen Insurance Partnerships

Prioritize contract renewals and expanded coverage agreements with the highest-revenue insurance providers. Simultaneously, explore partnerships with underrepresented insurers to diversify revenue streams and reduce dependency on a small number of payers.

Invest in Preventive Healthcare Programs

Given that middle-aged patients (35–54) drive the highest admission volumes, invest in preventive care initiatives — chronic disease management, wellness screenings, and lifestyle programs — to reduce avoidable hospitalizations and lower long-term costs.

Monitor High-Cost Hospitals Closely

Implement regular performance reviews for hospitals with above-average billing and length of stay. Investigate root causes — whether clinical complexity, operational inefficiency, or resource overutilization — and introduce standardized cost-control protocols.

Reduce Patient Length of Stay

Establish evidence-based discharge protocols and care pathway standards to minimize unnecessary hospital stays. Shorter stays improve bed availability, reduce costs, and lower the risk of hospital-acquired complications — directly benefiting both patients and the organization.

Build Real-Time Operational Dashboards

Extend the current Power BI dashboard into a real-time operational command center. Integrate live data feeds for admissions, bed occupancy, and billing to enable proactive decision-making rather than reactive reporting.

Skills Demonstrated

This project showcases a comprehensive skill set spanning technical data analytics and strategic business communication — the combination that defines a high-impact data professional in the healthcare domain.

Technical Skills

SQL

Advanced querying, aggregations, window functions, CASE logic, and business intelligence queries on structured healthcare data.

Python

Data cleaning pipelines, feature engineering, EDA, correlation analysis, and outlier detection using Pandas, NumPy, and Seaborn.

Power BI

Interactive dashboard design, DAX measures, Power Query transformations, and executive-level report development.

Data Cleaning & EDA

End-to-end data preparation including NULL handling, deduplication, type standardization, and exploratory visualization.

Business & Analytical Skills

Business Analysis

Translating raw data findings into clear, actionable business recommendations aligned with hospital management priorities.

KPI Development

Defining and computing key performance indicators — revenue, admissions, length of stay, and billing averages — that matter to stakeholders.

Data Storytelling

Structuring analytical findings into a coherent narrative that drives understanding and supports executive decision-making.

Executive Reporting

Designing dashboards and reports tailored for non-technical audiences, focusing on clarity, visual hierarchy, and insight density.

Problem Solving

Applying structured analytical thinking to diagnose operational issues and propose data-backed solutions for hospital management.

Project Summary & Conclusion

This project demonstrates a complete, end-to-end data analytics workflow — from raw, unstructured healthcare data to polished executive reporting. By combining SQL for data extraction and transformation, Python for exploratory analysis and statistical profiling, and Power BI for interactive visualization, this project delivers a comprehensive analytical solution that hospital leadership can use to monitor performance, identify trends, and make informed operational and financial decisions.

What Was Built

A fully functional analytics pipeline covering data cleaning, SQL querying, Python EDA, and an interactive Power BI executive dashboard with KPIs, trend visuals, and demographic breakdowns.

What It Enables

Hospital management can now monitor operational performance, track financial metrics, analyze patient demographics, and identify healthcare trends — all from a single, unified dashboard.

What It Proves

This project validates the ability to translate complex healthcare data into clear, actionable business intelligence — a critical competency for data analysts in the healthcare and consulting sectors.

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View the Full Project

All code, queries, and dashboard files are available on GitHub. Explore the complete project repository to review the analytical methodology and reproduce the findings.

View on GitHub

Thank you for reviewing this project. For questions or collaboration opportunities, please connect via GitHub or LinkedIn.